

Research Interests

I am interested in understanding the effect of super-massive black holes on their own growth and their larger surroundings. My work has focused on Broad Absorption Line Quasars (BALQSOs) which contain high velocity winds which can sometimes reach across their whole galaxy. These sources contain enhanced radio emission which is currently without a determined physical origin. I study these objects both on a large survey scale (creating the largest radio-detected sample of BALQSOs) and using VLBI high-resolution imaging to answer this origin question. This will have implications for many other black hole systems which show increased radio emission related to their accretion properties and dust content.

Research Experience

- 2020–Current **PhD Studies - High resolution imaging and survey studies of quasars with outflows using the LOFAR International Telescope**, *Durham University*, Supervisor - Dr Leah Morabito.
I study accreting black holes using the LOFAR telescope, a radio array spread across the whole of Europe. I began this work by using a wide field radio survey combined with spectroscopic imaging to show that the quasars that are radio-detected have different outflow properties as seen in their spectral absorption profiles. Now I am moving on to high resolution studies of these same systems. The processing of this data involves large super computers as even a small dataset for studying one particular object can be tens of terabytes. Work being led by my supervisor that I am now involved in was recently featured by the BBC - <https://www.bbc.co.uk/news/science-environment-57998940>.
- 2019–2020 **Investigating the Most Luminous Starbursts in the Universe**, *Durham University*, Supervisors - Prof. Ian Smail and Prof. Mark Swinbank.
This project combined data from several galaxy surveys in an attempt to understand the galaxy formation processes and histories of very bright, high-redshift and dusty galaxies. By matching galaxies across a large range of wavelengths and detectors, it is possible to determine many parameters of the galaxies using code such as MAGPHYS FORTRAN (da Cunha et al. 2008). This project required extensive use of Python for data analysis and visualisation. It also required understanding of how each galaxy survey was conducted and investigation into how to deal with non-detections, limits and issues with each of the surveys.
- 2019 **How viable are public contributions to asteroid astronomy?**, *Durham University*, Supervisors - Prof. Alastair Edge and Prof. Mark Swinbank.
This project used telescopes on the roof of the Durham Physics building to track several asteroids over two months. I used Python scripts to make calls to star catalogues and NASA's JPL HORIZONS system so that the location and identification of the asteroid in the image could be achieved automatically. I considered how this could be extended to a community effort and whether amateur-led teamwork could compete with NASA systems.

Publications

First Author

- 1 **Petley, J. W., Morabito, L. K., Alexander, D. M. and 6 colleagues**, *Connecting radio emission to AGN wind properties with Broad Absorption Line Quasars*, MNRAS, 525.4, 5159-5174, **2022**

Co-Author

- 1 Andonie, C. A., Alexander, D. M., Rosario, D. J and 13 colleagues, *A panchromatic view of infrared quasars: excess star formation and radio emission in the most heavily obscured systems*, MNRAS, 517.2, 2577-2598, **2022**

Telescope Experience

- LOFAR Member of LOFAR Surveys Key Science Project
e-MERLIN PI for 120 hours observed in Cycle 13

Public Engagement/Outreach

- Celebrate Science Helped children ages 6-16 understand cosmology work in Durham through exploring a "Universe Maker" and virtual reality tours of our simulations.
- Space Camp Overnight camp for children aged 10 at a local school. Created model planets and moon dust before leading a stargazing session.

Public Talks **Mexborough and Swinton Astronomical Society, Sep. 2022** - *How super-massive black holes control the galactic weather*

Scientific Presentations

South Africa, Dec. 2022 **South African Radio Astronomy Observatory** - Invited Speaker - *Determining the origin of radio emission in quasars with outflows*

South Africa, Dec. 2022 **University of Cape Town** - Invited Lunchtime Speaker - *Determining the origin of radio emission in quasars with outflows*

Iceland, Oct. 2022 **What Drives the Growth of Black Holes? Conference** - Contributed - *How can we determine the source of the enhanced radio detection fraction of quasars with strong winds?*

Virtual, Oct. 2022 **LOFAR Early Careers Researcher Conference** - Contributed - *How can we determine the source of the enhanced radio detection fraction of quasars with strong winds?*

Virtual, May 2022 **Kapteyn Institute** - Invited Lunchtime Speaker - *How can we determine the source of the enhanced radio detection fraction of quasars with strong winds?*

Virtual, Nov. 2021 **Science at Low Frequencies** - Contributed Lightning Talk

Education

2016–Current **Physics and Astronomy, MPhys, Durham University, Durham, 1st Class with Honours.**
Baxter Prize from Hatfield College in 2017 and 2018 for Academic Achievement

2009–2016 **Secondary School, King Edward VI Aston School, Birmingham, State Grammar School in Central Birmingham.**

Qualifications

A-Levels 2016
◦ Further Mathematics = A^* , Mathematics = A^* , Physics = A

AS-Levels 2015
◦ French = B

AS-Levels 2014
◦ Philosophy and Ethics = A

GCSEs 2014
◦ Mathematics, Physics, Chemistry, Biology, English Literature, English Language, French, Religious Studies = A^*
◦ History, Music = A

Programming Experience

Python Proficient in Python for Data Analysis and Physics Problem Solving. I also have some experience in web development with Django and Flask. I have previously won prizes at Durham Hackathon for a Flask MicroLending Service and a Fake News Detecting Whatsapp Bot each created in 24 hours.

C++ I took a long online course on the language in 2018 but have not had the chance to use it in a practical or work setting since

Machine Learning I have a small amount of machine learning experience mostly through experimentation with Google Keras and TensorFlow. This is an area I would like to learn a lot more about over the next few years.

Others I have been learning R and Julia over the past year, primarily in the context of data science and machine learning.

Other Experience

Paid

2019 **Summer Internship, Drone Industries Ltd., Gateshead, 7 weeks.**
I won the chance to work for a company in the space or space technologies industry through a Santander scheme. I was mostly given tasks related to website design and structure but I managed to implement some extra features such as interactive maps combined with drone imagery using javascript. I also used OpenCV to help identify different types of moor plants and study their growth in drone images for the local council. I combined artificial intelligence AWS services with archive images from a local museum to pull out useful text and key aspects of their extensive collection.

2016–2020 **Tutor**, *Advantage 11+*, Birmingham.

I have been tutoring children between the ages of 8 and 11 for several years. This has allowed me to:

- As a leader, manage a class of 25 young students and two other tutors
- Plan series of lessons between 2 and 4 weeks long
- Deliver short lessons on key topics in a clear and effective manner
- Communicate well with parents, giving advice on what their children need to work on and how they should be studying

2016–Current **Musician**, Durham.

I have played in several bands while at Durham University. I play clarinet (Grade 8 distinction), saxophones, piano (grade 7 merit) and I am learning the flute. I have been able to:

- Negotiate prices and payment with large university organisations for balls and other events
- Lead a big band of 18 people as president. Manage several thousand pounds for a year as well as providing leadership and direction.
- Organise a tour for 16 people to travel to Amsterdam. Liaise with transport and accommodation providers, set up venues to perform abroad and budget based on income for the year.
- Record two albums with a band in Durham and release them online.

Volunteering

2014 **Teaching Assistant**, *Wilson Stuart School*, Birmingham.

I volunteered as a teaching assistant at this Special Academy for disabled students. I worked with a class of students of very different abilities and needs. I improved:

- My confidence in responding to stressful or unusual situations (e.g. student having a seizure)
- Communication skills with the students and also teachers. I even learnt some "makaton" sign language
- Teaching skills. How to teach a 14-year-old student practical life skills when they struggle to count past 10.

Languages

French Moderate

B at AS Level

Further Achievements

Duke of Edinburgh Award I have completed my Gold Duke of Edinburgh award which involved many hours of volunteering, sporting activity and new experiences. Prince Edward presented me with the award in 2017.

Postgraduate Events Officer During the COVID-19 pandemic I was elected to the role of Events Officer for post-grads in my college. I ran successful online events both socially and for members to share their research with others across disciplines. By the end of the role I managed to run some in-person events, the highlight being a postgrad BBQ celebration.

Running Despite Adversity I was diagnosed with Juvenile Arthritis, a genetic disease which affects my joints, when I was 15. Despite this, I have been able to continue sporting activity and, in particular, running where I have been able to run half marathons (starting 2014) and the Budapest marathon (2018). I also play, and regularly captain teams in, Ultimate Frisbee for Durham University along with football for my college.